

Quantum Mechanics III

Lecture 9

Correlation Experiments

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$E(a, b) = \sum_{x,y=\pm 1} xyp(x, y|a, b)$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b')$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$|S| \leq 2 \text{ for registered factorized models}$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 9; C95-26 protocol; Correlation Registry CR-4